

EXPERIMENT 19:

Experiment: Naïve Bayes Classification for Bank Loan Prediction

```
print("J THASLIMA NASREEN/192424152")
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.naive_bayes import GaussianNB
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
```

Create bank loan dataset

```
data = {  
    'Age': [25, 35, 45, 28, 50, 40, 30, 55, 42, 29, 48, 33, 52, 27, 38, 46, 31, 57, 36, 44],  
    'Income': [25000, 50000, 80000, 30000, 90000, 65000, 40000, 95000, 70000, 28000,  
               85000, 45000, 100000, 32000, 55000, 75000, 42000, 110000, 60000, 72000],  
    'LoanAmount': [10000, 20000, 30000, 15000, 25000, 22000, 18000, 30000, 20000, 12000,  
                   28000, 18000, 35000, 14000, 19000, 25000, 17000, 40000, 21000, 26000],  
    'CreditScore': [600, 700, 750, 620, 780, 720, 650, 800, 730, 610,  
                    770, 680, 810, 630, 690, 740, 660, 820, 710, 760],  
    'Loan_Status': ['No', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'No',  
                    'Yes', 'No', 'Yes', 'No', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'Yes']  
}
```

```
df = pd.DataFrame(data)
```

Display dataset

```
print("Bank Loan Dataset:")
```

```
print(df)
```

Encode target variable

```
encoder = LabelEncoder()
```

```
df['Loan_Status'] = encoder.fit_transform(df['Loan_Status'])
```

```

# Separate input and output
X = df[['Age', 'Income', 'LoanAmount', 'CreditScore']]
y = df['Loan_Status']

Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Create Naïve Bayes classifier
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Display actual and predicted values
print("\nActual Values:")
print(y_test.values)

print("\nPredicted Values:")
print(y_pred)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

# Generate confusion matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")

```

```

print(cm)

# Display confusion matrix

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=encoder.classes_
)

disp.plot(cmap="Blues")

plt.title("Naïve Bayes - Bank Loan Prediction")

plt.xlabel("Predicted Loan Status")

plt.ylabel("Actual Loan Status")

plt.show()

```

OUTPUT:

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Bank Loan Dataset:

	Age	Income	LoanAmount	CreditScore	Loan_Status
0	25	25000	10000	600	No
1	35	50000	20000	700	Yes
2	45	80000	30000	750	Yes
3	28	30000	15000	620	No
4	50	90000	25000	780	Yes
5	40	65000	22000	720	Yes
6	30	40000	18000	650	No
7	55	95000	30000	800	Yes
8	42	70000	20000	730	Yes
9	29	28000	12000	610	No
10	48	85000	28000	770	Yes
11	33	45000	18000	680	No
12	52	100000	35000	810	Yes
13	27	32000	14000	630	No

14	38	55000	19000	690	Yes
15	46	75000	25000	740	Yes
16	31	42000	17000	660	No
17	57	110000	40000	820	Yes
18	36	60000	21000	710	Yes
19	44	72000	26000	760	Yes

Actual Values:

[1 0 1 1]

Predicted Values:

[1 0 1 1]

Accuracy: 1.0

Accuracy Percentage: 100.0 %

Confusion Matrix:

[[1 0]

[0 3]]

